

RETAIL SALES PREDICTION

Predicting Product Sales Using Machine Learning

A Data Science Approach to Retail Inventory Optimization

Data Sprint 2026 | Big Mart Sales Dataset

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Built with Altair AI Studio (RapidMiner)

PROBLEM STATEMENT

Why Sales Prediction Matters

Retail stores like Big Mart stock thousands of products across multiple outlets, but predicting exact sales for each product is difficult due to varying factors such as product type, pricing, and store characteristics.

Inaccurate forecasting leads to overstocking (wasted inventory, storage costs) or understocking (lost sales, unhappy customers).

Overstocking

Wasted inventory & storage costs

Understocking

Lost sales & unhappy customers

OBJECTIVE

To predict the sales of a product at a given outlet using historical sales data and product/outlet attributes, enabling better inventory and stocking decisions.

DATASET

Big Mart Sales Dataset

A real-world retail dataset containing product and outlet-level sales records, sourced from Kaggle (Big Mart Sales Prediction Datasets).

8,523

Training Rows

11

Feature Columns

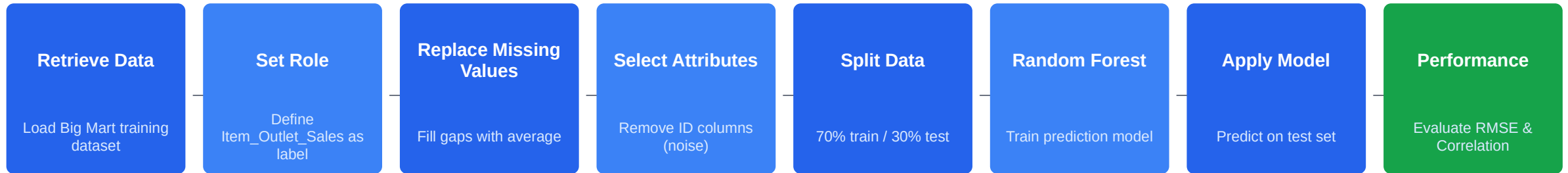
1

Target Variable

KEY COLUMNS

Column	Description	Column	Description
Item_Weight	Product weight	Outlet_Size	Small / Medium / High
Item_Fat_Content	Low Fat / Regular	Outlet_Location_Type	Tier 1 / 2 / 3
Item_Visibility	Display area allocated	Outlet_Type	Grocery Store / Supermarket
Item_MRP	Maximum retail price	Outlet_Establishment_Year	Year outlet was set up
Item_Type	Product category	Item_Outlet_Sales	TARGET — sales value

Workflow Design (RapidMiner / Altair AI Studio)



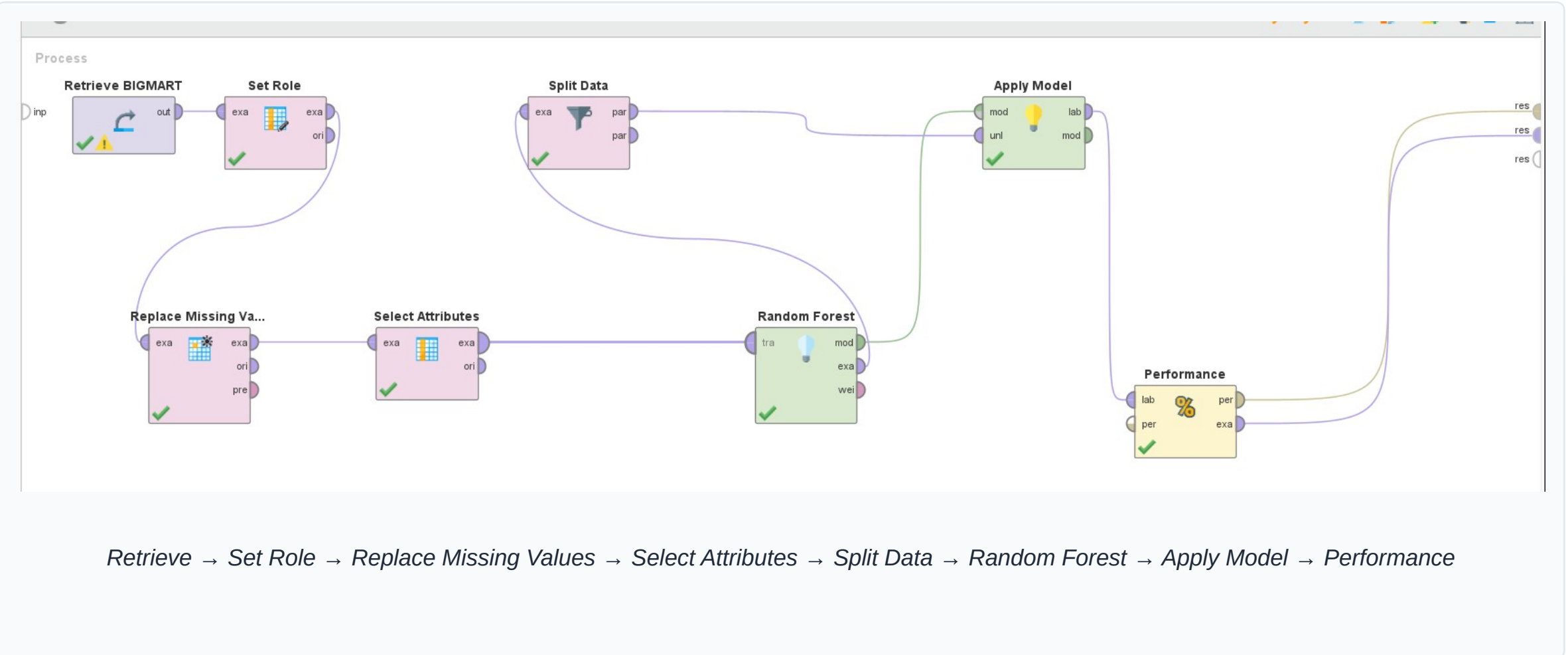
PIPELINE SUMMARY

Preprocessing: Missing value imputation + removal of non-predictive ID columns

Model: Random Forest Regression (100 trees, least-square criterion)

Validation: 70/30 train-test split, evaluated using RMSE and Correlation

Actual Workflow — Altair AI Studio (RapidMiner)



RESULTS

Model Performance

CORRELATION

0.841

Correlation between predicted and actual sales

ROOT MEAN SQUARED ERROR

1,030.13

Average prediction error in sales value

MODEL COMPARISON

Decision Tree

0.664

Gradient Boosted Trees

0.721

Random Forest (Final)

0.841

CONCLUSION

Key Takeaways

- A Random Forest regression model was built to predict product sales at retail outlets, achieving a correlation of 0.841 with actual sales.
- Removing non-predictive ID columns and handling missing values meaningfully improved model accuracy.
- The model can help retailers like Big Mart make data-driven inventory decisions — reducing overstock and stockouts.
- Future work: incorporating time-series seasonality and additional store-level features could further improve accuracy.

Thank You